



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SASTHA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

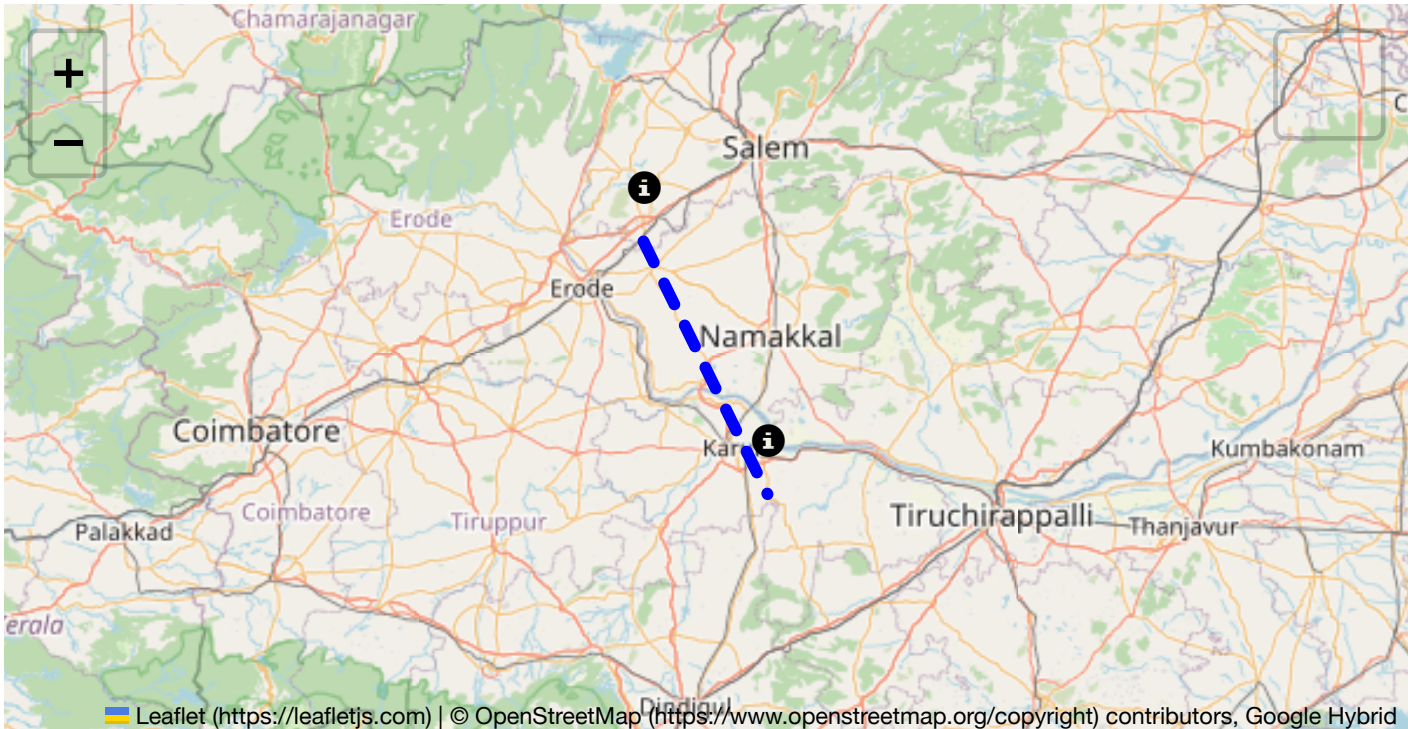
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

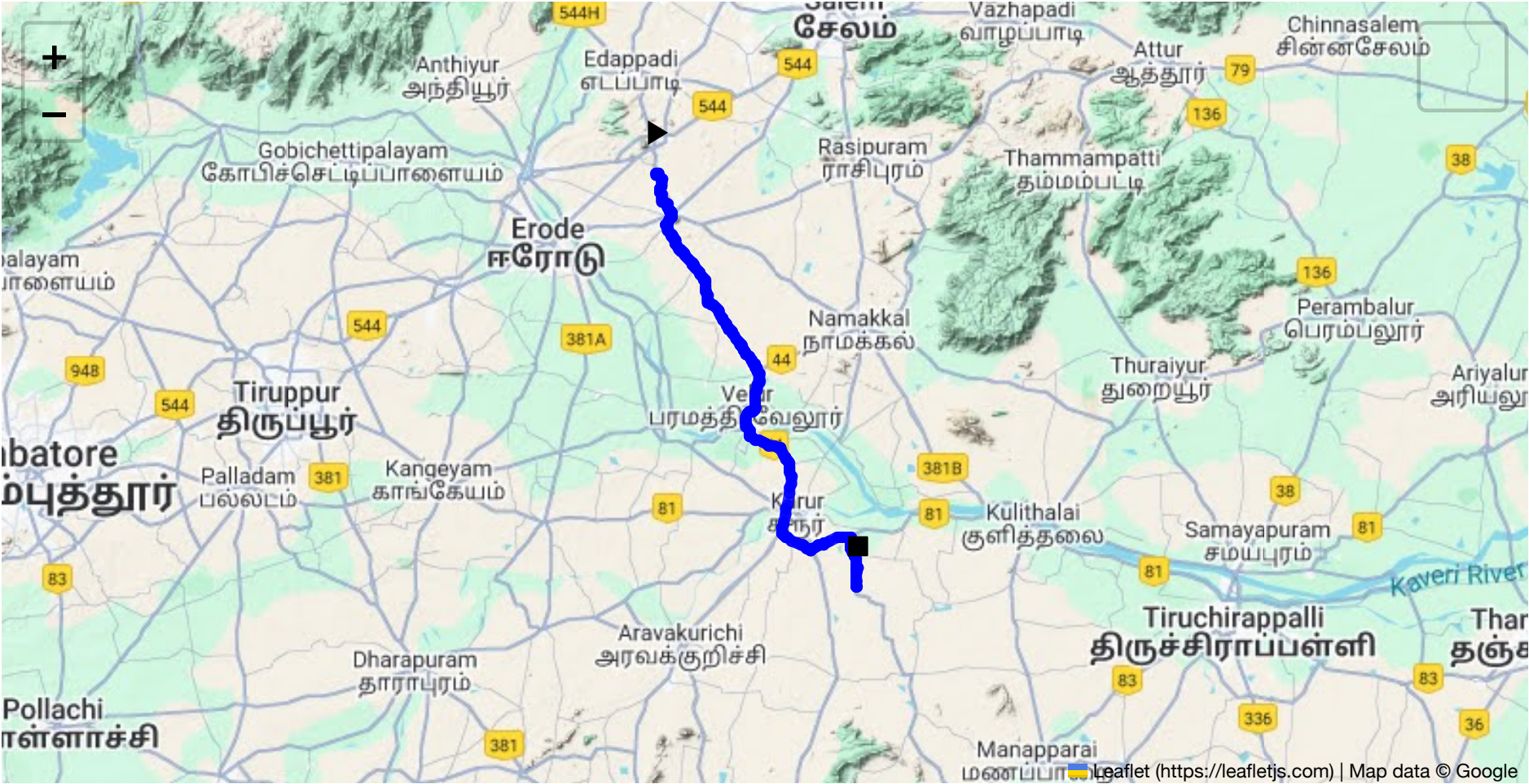
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 88.39 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours

Start: (11.4381, 77.8734)
End: (10.853312, 78.161736)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu, to V536+8M Vellianai North, Tamil Nadu, covers approximately 88.39 kilometers. The journey typically takes about 1.72 hours for heavy vehicles carrying hazardous materials. It traverses through a largely rural setting with some urban intersections and is a mix of national highways and state roads.

Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate with hot summers, significant monsoon rains from June to September, and mild winters. The monsoon season can lead to flooded roads and reduced visibility, especially in rural or low-lying areas along the route. Winds from cyclonic systems may also cause disruptions.

Traffic Patterns and Congestion

Traffic congestion is not as pronounced in rural Tamil Nadu compared to urban centers. However, Sangagiri and other towns along the route can experience moderate traffic during peak times, typically around 8 AM-10 AM and 5 PM-7 PM. The road near schools or market areas may see higher foot traffic and possible congestion.

Road Quality and Infrastructure

The National Highway segments are generally well-maintained, though some state highways might have potholes and narrower lanes. Watch for abrupt changes in road conditions when moving from national to state roads. Infrastructure quality can vary, with some areas lacking proper signage or well-defined shoulders.

Alternative Routes for Emergencies

- **Via Tiruchengode:** One alternative is heading via SH79 towards Tiruchengode and then reconnecting to the primary route. This route is longer but can serve as a detour during emergencies or roadworks.
- **Via Erode and Karur:** Another alternative goes through Erode and then connects with Karur. This path might provide better infrastructure in certain weather conditions.

Local Regulations on Hazardous Material Transport

Tamil Nadu mandates permitting for the transport of hazardous materials. Check official Tamil Nadu and national guidelines for specific road restrictions and required documentation. There are also time restrictions on when hazardous materials can be transported through certain areas.

Historical Incidents

There have been few notable incidents involving heavy vehicles or hazardous materials specifically along this route. General awareness of road conditions and adherence to transport regulations have minimized major accidents, but black spots identified by the local governments should be regularly monitored.

Environmental Considerations

Parts of the route traverse agricultural land and areas near water bodies. It's important to minimize any spill risks for hazardous materials to prevent environmental damage. Drivers should be aware of these sensitive areas, particularly around water bodies and near rural communities dependent on agriculture.

Communication Coverage

The majority of the route should have mobile coverage, but some rural areas might experience connectivity issues. It's advisable to account for these potential dead zones by preparing with offline maps and communication alternatives like satellite phones or radios.

Estimated Emergency Response Times

Emergency response in rural segments may be delayed because of the distance from major towns. Near urban areas, response times (for medical or fire emergencies) can range between 15-30 minutes, whereas rural sectors may experience times exceeding 45 minutes.

Risk Assessment Summary

Overall, this route is moderately safe for transporting hazardous materials, with the primary risks associated with weather conditions and potential rural road quality degradation. Preparedness in handling communications, awareness of environmental concerns, and knowledge of emergency procedures are essential. Drivers should adhere strictly to transportation regulations for hazardous materials and remain vigilant of potential traffic congestion in more populated towns. Regular updates on weather forecasts and road conditions are advisable to mitigate risks effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr
12	Turn	High	10.92509, 78.15310	15 KM/Hr
13	Turn	Medium	10.90842, 78.15408	30 KM/Hr
14	Turn	High	10.90229, 78.15716	15 KM/Hr
15	Turn	High	10.90225, 78.15799	15 KM/Hr
16	Turn	Medium	10.88119, 78.16362	30 KM/Hr
17	Blind Spot	Blind Spot	10.85366, 78.16142	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
19	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
20	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.92958, 78.05452



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.92509, 78.15310



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90842, 78.15408



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.90229, 78.15716



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.90225, 78.15799



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.88119, 78.16362



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.85366, 78.16142
